Ashish Sukumar

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Professional Summary

Robotics Engineering graduate student specializing in **motion planning, behavior planning, and trajectory optimization** for autonomous systems. Experienced implementing sampling-based and kinodynamic planners from scratch in **C++ and Python**, with hands-on work across high-dimensional configuration spaces, collision-aware planning, and behavior validation in simulation. Strong foundations in **kinematics, linear algebra, computational geometry, and optimization**. Seeking Summer/Fall 2026 internship to contribute to production-level autonomous vehicle planning systems and deploy algorithms on real-world platforms.

Education

Worcester Polytechnic Institute — **Massachusetts, USA**
M.S. Robotics Engineering — *2025–Present*
GPA: 4.00 / 4.00   Expected graduation: 2027
Relevant Coursework: Motion Planning, Foundations of Robotics, Deep Learning for Robotics, Computer Vision (RBE 549)

SRM Institute of Science and Technology — **Chennai, India**
B.Tech Computer Science and Engineering — *2021–2025*
CGPA: 9.54 / 10

Technical Skills

Languages: C++ (primary), Python
Motion Planning: RRT, RRT*, PRM, PRM*, RRTConnect, AO-RRT, KPIECE1, EST, SST, PDDL / TAMP
Planning Frameworks: OMPL, MoveIt
Trajectory & Control: Kinodynamic Planning, Trajectory Optimization, Feedback Control, Dynamic Systems Modeling, Controller Tuning
Kinematics & Math: Forward/Inverse Kinematics, Linear Algebra, Calculus, Computational Geometry, Collision Detection
Autonomous Systems: Behavior Planning, Decision-Making, State Machine Design, Simulation-to-Real Deployment
Simulation: Webots, Gazebo, Genesis Simulator
Robotics Stack: ROS2, MoveIt, OMPL
Hardware Platforms: Arduino (Uno, Nano, Nano 33 BLE Sense), ESP32, Raspberry Pi
Sensors: IMU, Gas, IR, Ultrasonic, Proximity, Camera Modules
3D Vision: NeRF, SfM, COLMAP, Pose Estimation, Point Cloud Processing, SLAM
Deep Learning: PyTorch, TensorFlow, CNNs, Spatial Transformer Networks
Tools: Linux, Git, CMake

Experience

WPI Small Business Digitization Initiative (SBDI) — **Worcester, MA**
Digital Consultant — *Oct 2025–Dec 2025*
- Worked independently to scope, design, and deliver solutions for small business clients end-to-end, managing full project lifecycle with minimal supervision.
- Applied iterative design and feedback workflows to produce client deliverables, developing structured communication and stakeholder collaboration skills.

Robocare Lab, Worcester Polytechnic Institute — **Worcester, MA**
Voluntary Research Assistant — *Oct 2025–Dec 2025*
- Contributed to robot **behavior design and perception pipelines** for the SoftBank **Pepper Robot**, integrating speech, gesture, and visual feedback modalities using **ROS2**.
- Configured and tested onboard camera, microphone, and joint actuator systems; identified and resolved behavioral failure modes through iterative lab testing.
- Supported experimental design, data collection, and evaluation of multimodal feedback systems in controlled environments.

e-Yantra (IIT Bombay) — **Chennai, India**
Junior Project Technical Assistant — *Jun 2024–Feb 2025*

- Designed and implemented **autonomous task pipelines** and simulation environments in **Webots Simulator** for the national eYSRC competition, integrating perception and **decision-making logic** end-to-end.
- Developed **embedded systems** software and sensor-based image processing modules enabling real-world robot environment interaction and autonomous behavior.
- Performed **system-level debugging** of hardware-software integration issues across sensor, actuation, and control layers.
- Formally recognised by Prof. Kavi Arya (Principal Investigator, IIT Bombay) for technical proficiency; delivered two workshops on Embedded Systems and Robotics to school students nationally.

Key Projects

Task and Motion Planning (TAMP) – 7-DOF Manipulator: Designed a full **TAMP pipeline** integrating **PDDL symbolic planning** with **OMPL motion planning**, enabling a 7-DOF robot arm to execute collision-aware motion primitives end-to-end. Implemented predicate extraction (ON, CLEAR, HOLDING, HANDEMPTY) and **closed-loop re-planning** for fault recovery. Demonstrates complete **behavior planning lifecycle**: task specification, planner selection, execution, and failure recovery – directly analogous to AV behavior planning systems.

High-Dimensional Sampling-Based Motion Planning (7-DOF) – C++ / OMPL: Implemented **PRM, PRM*, and RRTConnect** planners in **7-DOF joint space** with a custom **self-collision checker** for articulated robots. Solved **narrow-passage planning problems** using targeted sampling strategies. Benchmarked planner performance on **path optimality and cycle time** – findings directly applicable to AV motion primitive selection and trajectory quality analysis.

Custom RRT Planner – Implemented from Scratch in C++: Implemented a complete **Rapidly-Exploring Random Tree (RRT)** planner from scratch in **C++** with geometry-aware **collision detection** for articulated robots up to 20 links in a **233-dimensional configuration space**. Demonstrates strong command of **computational geometry, kinematics, and production-level C++** – directly applicable to Kodiak's planning codebase.

Kinodynamic Motion Planning for Constrained Systems – AO-RRT: Implemented **AO-RRT** for dynamically constrained robot models including **car-like robots** and pendulum systems – directly analogous to AV kinodynamic constraints (curvature limits, velocity bounds, turning radius). Benchmarked **KPIECE1, EST, and SST** for convergence speed and path quality across different system dynamics. Findings applicable to **controller selection and trajectory optimization** for real autonomous driving systems.

Fire Aware Smart-Bot – Autonomous Navigation and Sensor Fusion: Designed and deployed an **autonomous navigation system** with real-time sensor fusion (IR, gas, proximity), **shortest-path planning**, obstacle avoidance, BLE-based mobile control, and live video streaming on a custom embedded platform. Performed iterative **hardware-software debugging** and validated system performance in real-world deployment. Presented at **ICIoT 2025**.

Structure from Motion (SfM) Pipeline – Multi-View 3D Reconstruction: Built a full incremental SfM pipeline: **SIFT** feature matching, **RANSAC**-based Fundamental matrix estimation, Essential matrix decomposition, triangulation, **PnP-RANSAC** camera registration, and **bundle adjustment**. Reconstructed **1,318 3D points** across 4 cameras from 5 real images – applicable to geometric map construction for autonomous vehicle localization.

Neural Radiance Fields (NeRF) – 3D Scene Reconstruction: Built a complete NeRF pipeline from scratch in **PyTorch** – positional encoding, hierarchical sampling, volume rendering – achieving **27.42 dB PSNR / 0.9084 SSIM**. Extended to a custom dataset via COLMAP-based SfM for pose extraction. Applicable to map building and scene understanding in autonomous driving perception stacks.

Publications

Ashish, S., Jeya, R., Rithish, R., Donipart, R., & Ganesh, K. (2025). *Fire Detection and Risk Prediction for Smart Safety: A Neural Network-Driven IoT Approach.* IJERCSE.

Ashish, S., Jeya, R., Rithish, R., Donipart, R., & Ganesh, K. (2025). *Fire Aware Smart-Bot with AI Responsive System.* ICIoT 2025.

Certifications & Achievements

- Red Hat Certified System Administrator (RHCSA)
- Oracle Cloud Infrastructure (OCI) Certified
- Excellence Award for Outstanding Student & Best Class Representative – SRMIST
- 2nd Place – Project Expo, SRMIST